

# Statistical Regression - 3

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# OUTLINE OF THE COURSE

- Simple Linear Regression
- Multiple Linear Regression
- Generalised Linear Models
- Nonlinear Regression
- Mixed Effects Models

## BOOK FOR PART 3

- Chen, D.-G. and Chen, J.K. (2021). Statistical Regression Modeling with R. Springer.
- Gelman, A. and Hill, J. (2007). J. Applied Regression and Multilevel Modeling. Cambridge University Press.
- Heumann, C, Schomaker, M. and Shalabh (2022). Introduction to Statistics and Data Analysis. Springer.
- Montgomery, D.C., Peck, E.A. and Vining G.G. (2012). Introduction to Linear Regression Analysis (Fifth Edition). Wiley.

## LOGISTIC REGRESSION

- The previous set of slides dealt with continuous variables but what about a response (*dependent variable*) taking only a finite number of integer values?
- Consider people applying for mortgages (or subject to surgery): are they able to pay the mortgage back (or will they survive)?
- The observations are 1's (pays back/survives) and 0's (does not pay back/dies)
- We are still interested in studying the effect of covariates (*independent variables*), like age and gender, on the final result
- We cannot use  $Y = \beta_1 + \beta_2 X + \epsilon$  with  $Y = 0, 1$  since it is almost impossible to choose r.h.s. terms such that there is always either 0 or 1 in the l.h.s.
- We consider  $\pi = P(Y = 1)$  but we cannot use  $\pi = \beta_1 + \beta_2 X + \epsilon$  since it is almost impossible to choose r.h.s. terms such that the l.h.s. will be always between 0 and 1
- (logit) transformation:  $\log \left( \frac{\pi}{1 - \pi} \right) = X' \beta$ , with  $X', \beta$  vectors of size  $k$
- In simple linear regression:  $X' = (1, X), \beta' = (\beta_1, \beta_2)$

# LOGISTIC REGRESSION

- $\log \left( \frac{\pi}{1 - \pi} \right) = X' \beta \Rightarrow \pi = \frac{e^{X' \beta}}{1 + e^{X' \beta}}$
- For each  $i = 1, \dots, n$ , consider  $n_i$  observations  $(y_i, x_i)$  and the related probability  $\pi_i$  (e.g.  $y_i$ , out of  $n_i$ , persons with features  $x_i$ , paid the mortgage back)
- $\underline{y} = (y_1, \dots, y_n)$ ,  $\underline{x} = (x_1, \dots, x_n)$ ,  $\underline{\pi} = (\pi_1, \dots, \pi_n)$  and  $\underline{n} = (n_1, \dots, n_n)$
- Binomial model (Bernoulli if  $n_i = 1$ ) for  $Y$  and not Gaussian as in linear regression

$$P(Y_i = y_i | \pi_i, n_i, x_i) = \binom{n_i}{y_i} \pi_i^{y_i} (1 - \pi_i)^{n_i - y_i} = \binom{n_i}{y_i} \left( \frac{e^{x_i' \beta}}{1 + e^{x_i' \beta}} \right)^{y_i} \left( \frac{1}{1 + e^{x_i' \beta}} \right)^{n_i - y_i}$$

- Likelihood:  $\prod_{i=1}^n \binom{n_i}{y_i} \frac{e^{y_i x_i' \beta}}{(1 + e^{x_i' \beta})^{n_i}}$
- Log-likelihood:  $\sum_{i=1}^n \log \binom{n_i}{y_i} + \sum_{i=1}^n y_i x_i' \beta - \sum_{i=1}^n n_i \log (1 + e^{x_i' \beta})$

# LOGISTIC REGRESSION

- Likelihood  $L(\beta)$ : 
$$\prod_{i=1}^n \binom{n_i}{y_i} \frac{e^{y_i x_i' \beta}}{(1 + e^{x_i' \beta})^{n_i}}$$
- Log-likelihood  $l(\beta)$ : 
$$\sum_{i=1}^n \log \binom{n_i}{y_i} + \sum_{i=1}^n y_i x_i' \beta - \sum_{i=1}^n n_i \log (1 + e^{x_i' \beta})$$
- The MLE  $\hat{\beta}$  cannot be obtained in closed form and numerical methods are needed to solve the optimisation problem  $\hat{\beta} = \arg \max_{\beta} l(\beta)$
- $\pi_i = \left( \frac{e^{x_i' \beta}}{1 + e^{x_i' \beta}} \right) \Rightarrow \hat{\pi}_i = \left( \frac{e^{x_i' \hat{\beta}}}{1 + e^{x_i' \hat{\beta}}} \right)$  estimated probability for items/population with features  $x_i$
- Prior distribution on  $\beta$ : e.g. Multivariate Gaussian (simplest: product of independent univariate Gaussian distributions)
- Neither posterior distribution in closed form nor full conditionals to use Gibbs sampling: conditional known up to a constant  $\Rightarrow$  Metropolis-Hastings algorithm

# LOGISTIC REGRESSION

- We saw in simple linear regression that  $\beta_0$  is the value when  $X = 0$  and  $\beta_1$  is the quantity by which  $Y$  increases when  $X$  is increased by 1
- Here the interpretation of the parameters is more complex since we have, e.g.

$$\log \left( \frac{\pi}{1 - \pi} \right) = \beta_0 + \beta_1 X \Rightarrow \pi = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$$

- We have to think of the odds:  $\frac{\pi}{1 - \pi} = e^{\beta_0 + \beta_1 X}$
- If we increase  $X$  by 1 we get new odds:  $\frac{\pi'}{1 - \pi'} = e^{\beta_0 + \beta_1 (X+1)} = e^{\beta_0 + \beta_1 X} e^{\beta_1}$
- $\Rightarrow$  Odds ratio  $O = \frac{\pi'}{1 - \pi'} \bigg/ \frac{\pi}{1 - \pi} = e^{\beta_1}$
- We can use odds ratio (between new and old model) to measure the influence of adding a new variable: if  $O \neq 1$  (better, not in  $(1 - \delta, 1 + \delta)$ ), then the variable has an influence on the probability  $\pi$  (positive if  $O \gg 1$  and negative if  $O \ll 1$ )

## LOGISTIC REGRESSION

- Goodness of fit of logistic regression assessed using deviance, comparing a model(\*) to a saturated one where any observation  $y_i$  (if all have different covariates) or group of  $n_i$  homogeneous observations (same covariates like here) has its own probability, estimated by  $y_i/n_i$  in a group with  $y_i$  1's out of  $n_i$  elements
- We say null deviance if compared to a model(\*) with just the intercept and residual deviance if compared with proposed model(\*)
- The estimates of the probabilities of the model(\*) are  $\hat{\pi}_i = e^{x_i'\hat{\beta}} / (1 + e^{x_i'\hat{\beta}})$
- Deviance: twice the difference in log-likelihoods between saturated and model(\*)

$$D = 2 \sum_{i=1}^n \left[ y_i \log \left( \frac{y_i}{n_i \hat{\pi}_i} \right) + (n_i - y_i) \log \left( \frac{n_i - y_i}{n_i (1 - \hat{\pi}_i)} \right) \right]$$

- Small values of  $D$  imply that model(\*) provides a satisfactory fit to the data, while large values of  $D$  imply that model(\*) is not adequate
- Goodness of fit test: hypothesis of good fit  $\Rightarrow$  both null and residual deviances should have  $\chi^2$  distributions (with  $n - 1$  and  $n - p - 1$  degrees of freedom, respectively)  
 $\Rightarrow$  hypothesis rejected for large values of the deviance



# LOGISTIC REGRESSION

- Survey of 3200 residents in a small area of Bangladesh suffering from arsenic contamination of groundwater\*
- Respondents with elevated arsenic levels in their wells were encouraged to switch their water source to a safe well in the nearby area and the survey was conducted several years later to learn which of the affected residents had switched wells
- The goal of the analysis is to learn about the factors associated with switching wells
- To start, we will use `dist` (the distance from the respondent's house to the nearest well with safe drinking water) as the only predictor of `switch` (1 if switched, 0 if not).
- Then we will expand the model by adding the arsenic level of the water in the resident's own well as a predictor and then we will add all variables
- After loading the `wells` data, we first rescale the `dist` variable (measured in meters) so that it is measured in units of 100 meters

\*Example due to Gabry and Goodrich (website), based on Gelman and Hill's book

# LOGISTIC REGRESSION

```
library(rstanarm)
data(wells)
wells$dist100 <- wells$dist / 100
head(wells)
library(ggplot2)
ggplot(wells, aes(x=dist100, y=after_stat(density), fill=switch==1)) +
  geom_histogram() + scale_fill_manual(values=c("gray30", "skyblue"))
```

- Distribution of dist100: 1737 residents who switched (blue bars) and 1283 who did not (dark grey bars)
- It is just one density (not two!) which describes also the proportion of switch/no switch at various distances
- For the residents who switched wells, the distribution of `dist100` is more concentrated at smaller distances
- We start presenting the frequentist analysis

## LOGISTIC REGRESSION

```
fitm1=glm(switch ~ dist100,data=wells, family = binomial(link="logit"))
summary(fitm1,digits=3)
fitm2=glm(switch ~ dist100 + arsenic),data=wells,
family = binomial(link="logit"))
round(coef(fitm2), 3)
summary(fitm2,digits=3)
fitm3<-glm(switch ~ arsenic+assoc+educ+dist100,data=wells,
family = binomial(link = "logit"))
summary(fit3,digits=3)
```

## LOGISTIC REGRESSION

We consider now a Bayesian approach with a Student  $t$  prior with coefficients close to 0 but with chances of being large (less likely under Gaussian)

```
t_prior <- student_t(df = 7, location = 0, scale = 2.5)
fit1<-stan_glm(switch ~ dist100,data=wells,seed = 12345,
  family = binomial(link = "logit"),
  prior = t_prior, prior_intercept = t_prior)
summary(fit1,digits=3)
round(posterior_interval(fit1, prob = 0.5), 3) # digits=3
fit2 <- update(fit1, formula = switch ~ dist100 + arsenic)
round(coef(fit2), 3)
summary(fit2,digits=3)
fit3<-stan_glm(switch ~ arsenic+assoc+educ+dist100,data=wells,
family = binomial(link = "logit"),seed = 12345,
  prior = t_prior, prior_intercept = t_prior)
summary(fit3,digits=3)
```

# LOGISTIC REGRESSION

- *Frequentist* and **Bayesian** estimates
- `switch` – binary/dummy (0 or 1) for well-switching
- *0.467* - **0.468**: `arsenic` – arsenic level in respondent's well
- *-0.896* - **-0.897**: `dist100` – distance (100 meters) from the respondent's house to the nearest well with safe drinking water
- *-0.124* - **-0.125**: `association` – binary/dummy (0 or 1) if member(s) of household participate in community organizations
- *0.042* - **0.043**: `educ` – years of education (head of household)
- Frequentist and Bayesian estimates almost the same: many data (3020) and no very concentrated prior

## LOGISTIC REGRESSION

- Using the coefficient estimates from the first model, we can plot the predicted probability of `switch = 1` (as a function of `dist100`)
- `plogis` is the cdf of a logistic distribution

```
t_prior <- student_t(df = 7, location = 0, scale = 2.5)
fit1<-stan_glm(switch ~ dist100,data=wells,seed = 12345,
  family = binomial(link = "logit"),
  prior = t_prior, prior_intercept = t_prior)
summary(fit1,digits=3)
pr_switch <- function(x, ests) plogis(ests[1] + ests[2] * x)
coef(fit1)[1]; coef(fit1)[2]
aa=seq(0,12,0.25)
plot(aa,pr_switch(aa,coef(fit1)),type='l')
```

## EXAMPLE: THEATER DATA \*

- Survey (`theatre.csv`) with 699 participants in a Swiss city, all frequent visitors to a theatre, asked about age, sex (female=1), annual income (in 1000 SFR), expenditure on cultural activities (in SFR per month), expenditure on theatre (in SFR per month), and estimated expenditure on theatre in the previous year (in SFR per month)
- Logistic regression model well integrated into R through command `glm`
- Same parameters as in linear regression (`formula` and `data`), and the additional family specified as `family=binomial(link='logit')` or `family=binomial`
- Theatre (expenditure per month is the selected response variable but it is not binary  
⇒ transform into 1 the expenditures > 150 SFR and 0 otherwise

```
theatre=read.csv("theatre.csv")
head(theatre)
theatre$Theatre.bin <- ifelse(theatre$Theatre>150,1,0)
attach(theatre)
model.log <- glm(Theatre.bin ~ Age+Sex+Income+Culture+ Theatre_ly,
family=binomial(link='logit'),data=theatre)
summary(model.log)
```

*\*From Heumann, Schomaker, Shalabh (2022), Introduction to Statistics and Data Analysis, Springer*

## EXAMPLE: THEATER DATA

```

                Estimate Std. Error z value Pr(>|z|)
(Intercept) -8.512305    0.845365 -10.069  < 2e-16 ***
Age          0.013594    0.007274   1.869   0.06164 .
Sex          0.634535    0.196549   3.228   0.00124 **
Income       0.039834    0.007830   5.087  3.63e-07 ***
Culture      0.016468    0.002177   7.564  3.90e-14 ***
Theatre_ly   0.002134    0.004219   0.506   0.61299
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

- As in linear regression, the output shows `Estimate`, `Std.Error`, `z value`, `Pr(> |z|)`, where *t*-statistics is replaced by a *z*-statistics based on the asymptotic normal distribution of the MLE
- The variables `Age`, `Sex`, `Income` and `Culture` have a significant influence on whether a visitor will spend more than 150 SFR per month (or not) if we choose a significance level of  $\alpha = 0.1$  (since their *p*-values are lower than  $\alpha$ )
- The coefficients are all positive (except the intercept) and the influence of each variable, keeping the other fixed, is given by the odds ratios

```
print(exp(coef(model.log)), digits=2)
```



## EXAMPLE: THEATER DATA

```
print(exp(coef(model.log)), digits=2)
(Intercept)      Age      Sex      Income  Culture  Theatre_ly
      0.0002    1.0137    1.8861    1.0406    1.0166    1.0021
```

- Odds for females ( $Sex=1$ ) to spend more than 150 SFR (versus less than or equal to 150 SFR) is  $\exp(0.634535) = 1.886145$  higher than the odds for males
- Visitors who are older or have a higher income, or have higher general expenditures for culture have higher odds of spending more than 150 SFR for a theatre visit
- For example, for each SFR that is spent more on general expenditures for `Culture`, the odds for spending more than 150 SFR for theatre visits is higher by a factor of  $\exp(0.016468) = 1.016604$ , i.e. 1.66% higher
- Unfortunately, the summary output of the logistic regression model contains only the coefficients and not the odds ratios themselves or the confidence intervals for the odds ratios

## EXAMPLE: THEATER DATA

- Fortunately, the function `confint` in the standard MASS package can be applied directly on a `glm` object, providing confidence intervals for the regression coefficients
- Taking the exponential of lower and upper bounds of these confidence intervals provides confidence intervals for the odds ratios, where the default is a 95%-confidence interval

```
exp(confint(model.log))
              2.5%   97.5%
(Intercept) 0.0000 0.0010
Age          0.9993 1.0283
Sex          1.2863 2.7822
Income       1.0250 1.0570
Culture      1.0124 1.0211
Theatre_ly   0.9939 1.0105
```

- Model selection performed using Akaike Information Criterion (AIC)

```
library(MASS); stepAIC(model.log)
```

## EXAMPLE: THEATER DATA

Start: AIC=743.17

Theatre.bin ~ Age + Sex + Income + Culture + Theatre\_ly

|              | Df | Deviance | AIC    |
|--------------|----|----------|--------|
| - Theatre_ly | 1  | 731.42   | 741.42 |
| <none>       |    | 731.17   | 743.17 |
| - Age        | 1  | 734.66   | 744.66 |
| - Sex        | 1  | 741.78   | 751.78 |
| - Income     | 1  | 758.46   | 768.46 |
| - Culture    | 1  | 802.78   | 812.78 |

Step: AIC=741.42

Theatre.bin ~ Age + Sex + Income + Culture

|           | Df | Deviance | AIC    |
|-----------|----|----------|--------|
| <none>    |    | 731.42   | 741.42 |
| - Age     | 1  | 735.51   | 743.51 |
| - Sex     | 1  | 741.89   | 749.89 |
| - Income  | 1  | 761.56   | 769.56 |
| - Culture | 1  | 803.23   | 811.23 |

Call: glm(formula = Theatre.bin ~ Age + Sex + Income + Culture, family = binomial(link = "logit"), data = theatre)

Coefficients:

| (Intercept) | Age     | Sex     | Income  | Culture |
|-------------|---------|---------|---------|---------|
| -8.31499    | 0.01437 | 0.62872 | 0.04071 | 0.01648 |

Degrees of Freedom: 698 Total (i.e. Null); 694 Residual

Null Deviance: 897.9

Residual Deviance: 731.4 AIC: 741.4

## EXAMPLE: THEATER DATA

```
stepAIC(model.log, direction="both")
```

- Backward elimination by default, but possible to specify `backward`, `forward` and `both`
- `none` denotes the AIC with all the parameters in the current model, while the other AICs are computed when adding or removing one variable
- Excluded variables are listed with a "+" and only in `forward` and `both` while variables in the current model are listed with a "-" and only in `backward` and `both`

```
Step:  AIC=741.42
```

```
Theatre.bin ~ Age + Sex + Income + Culture
```

|              | Df | Deviance | AIC    |
|--------------|----|----------|--------|
| <none>       |    | 731.42   | 741.42 |
| + Theatre_ly | 1  | 731.17   | 743.17 |
| - Age        | 1  | 735.51   | 743.51 |
| - Sex        | 1  | 741.89   | 749.89 |
| - Income     | 1  | 761.56   | 769.56 |
| - Culture    | 1  | 803.23   | 811.23 |

# GENERALISED LINEAR MODEL

- In linear regression we saw  $Y$  with Gaussian distribution, while in logistic regression it had a Bernoulli distribution
- It is possible to consider other distributions for  $Y$  (in the so called exponential family), like Poisson, binomial, exponential and gamma
- The key idea is to model  $E(Y)$  as a function of  $X'\beta$
- In logistic regression  $E(Y) = \pi$  and we used the logit function to get the logistic function  $\pi = \frac{e^{\beta_0 + \beta_1 X}}{1 + e^{\beta_0 + \beta_1 X}}$
- The logistic function is an example of a `link function`, which is linking  $E(Y)$  and  $X'\beta$ , and other link functions characterise the cases of Poisson and the other distributions
- The models with these characteristics are called `Generalised Linear Models` (GLMs)
- We will see another example of GLM: Poisson regression

# GENERALISED LINEAR MODEL

- We think of the GLM as  $Y = EY + \varepsilon$ , with  $EY = \mu$  and an error dependent on the model
- Considering covariates, the observations are  $(y_i, \mathbf{x}_i)$  with  $\mathbf{x}_i' = (1, x_{i1}, \dots, x_{ip})$ ,  $i = 1, \dots, n$ , such that  $E(y_i) = \mu_i$
- In GLMs we connect  $\mu_i$  to the linear predictor  $\mathbf{x}_i'\boldsymbol{\beta}$  via a link function  $g(\cdot)$   
$$\Rightarrow g(\mu_i) = \eta_i = \beta_0 + \beta_1 x_{i1} + \dots + \beta_p x_{ip} = \mathbf{x}_i'\boldsymbol{\beta} \Rightarrow \mu_i = g^{-1}(\eta_i) = g^{-1}(\mathbf{x}_i'\boldsymbol{\beta})$$
- Linear regression is obtained considering the identity link:  $g(\mu_i) = \mu_i = \mathbf{x}_i'\boldsymbol{\beta}$
- The log link  $g(\mu_i) = \log(\mu_i) = \mathbf{x}_i'\boldsymbol{\beta} \Rightarrow$  Poisson regression
- Poisson regression is used when  $Y$  represents counts of relatively rare events, like detection of bugs in software testing or gas escapes, and there is interest in linking the rate of those events to some features (covariates)

## POISSON REGRESSION

- We consider a response (r.v.)  $Y$  taking integer values with a Poisson distribution

$$P(Y = y) = \frac{\mu^y}{y!} e^{-\mu}, \text{ with } \mu > 0$$

- It is possible to show that  $E(Y) = \mu$  and  $Var(Y) = \mu$

- $g(\mu_i) = \log(\mu_i) = \mathbf{x}_i' \boldsymbol{\beta} \Rightarrow \mu_i = e^{\mathbf{x}_i' \boldsymbol{\beta}}$

- Consider  $n$  observations  $(y_i, \mathbf{x}_i'), i = 1, \dots, n$

- Likelihood

$$\begin{aligned} L(\mathbf{y}, \boldsymbol{\beta}) &= \prod_{i=1}^n f_i(y_i) = \prod_{i=1}^n \frac{\mu_i^{y_i} e^{-\mu_i}}{y_i!} = \frac{\prod_{i=1}^n \mu_i^{y_i} e^{-\sum_{i=1}^n \mu_i}}{\prod_{i=1}^n y_i!} \\ &= \frac{\prod_{i=1}^n e^{y_i \mathbf{x}_i' \boldsymbol{\beta}} e^{-\sum_{i=1}^n e^{\mathbf{x}_i' \boldsymbol{\beta}}}}{\prod_{i=1}^n y_i!} \end{aligned}$$

- $\hat{\boldsymbol{\beta}}$  obtained numerically (Note: Number of Fisher Scoring iterations: in output for `summary`: related to iterations to get stable value of MLE)

## POISSON REGRESSION\*

- Goal: Factors affecting number of papers by students in biochemistry PhD programs
- Variables:
  - `art`: papers in last 3 years of PhD (target, i.e.  $Y$ )
  - `fem`: gender of student
  - `mar`: marital status of student
  - `kid5`: number of children aged 5 or younger
  - `phd`: prestige of PhD department
  - `ment`: papers by PhD mentor during last 3 years

```
library(pscl)
data(bioChemists)
dim(bioChemists)
head(bioChemists)
summary(bioChemists)
hist(bioChemists$art, xlab="Articles Published", main="")
```

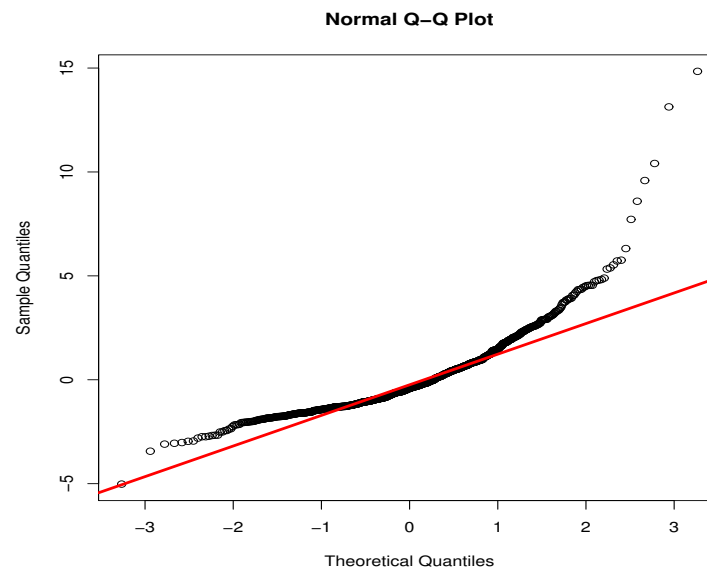
\*From Chen and Chen (2021), Statistical Regression Modeling with R, Springer.



# POISSON REGRESSION

- We try first linear regression but Q-Q suggests it is not appropriate

```
art.lm =lm(art~., bioChemists)
summary(art.lm)
qqnorm(art.lm$residuals)
qqline(art.lm$residuals, lwd=3, col="red")
```



# POISSON REGRESSION

- Linear regression is not appropriate not only because of Q-Q plot but  $Y$ , i.e. art, non negative and integer  $\Rightarrow$  try Poisson regression
- We get some significant variables but the residual deviance, testing the null hypothesis that the proposed model fits the data, has a  $\chi^2$  distribution s.t. the  $p$ -value  $< 0.0001 \Rightarrow$  the model does not fit well
- Furthermore, variance and mean in a Poisson model should be equal but here their ratio is 2.19
- Chen and Chen propose quasi-Poisson regression or negative binomial regression

```
art.poisson = glm(art~., data=bioChemists, family =c("poisson"))
summary(art.poisson)
pval.poisson= 1-pchisq(deviance(art.poisson),df.residual(art.poisson))
pval.poisson
var.art = var(bioChemists$art); var.art
mean.art = mean(bioChemists$art); mean.art
var.art/mean.art
```